Beyond Forecast Accuracy: Feasibility-Constrained Demand Planning

Yihua Xu
Independent Researcher
San Jose, California, USA
yxu442@gmail.com

Abstract

Demand planning teams optimize forecast accuracy; operations teams execute plans. These objectives diverge when an accurate forecast requires plan changes that exceed operational absorption capacity. This paper reframes forecast-driven demand planning as feasibility-constrained model selection through a forecast-to-plan feasibility layer. Candidate forecasts are converted into executable planning signals, measured for inventory exposure, plan stability, execution violations, and governance burden, and ranked by feasibility-adjusted planning utility rather than accuracy alone.

The empirical design uses DataCo as an execution-risk anchor and three demand-planning datasets: Favorita for the main proof of concept, M5 for hierarchy and intermittency, and Walmart for secondary business-context and weekly-cadence robustness. Across 123 paper-facing scenario and robustness evaluation groups, the accuracy-first deployable strategy differs from the operational-loss-optimal deployable strategy in 91.1% of cases. The accuracy-first planning-execution gap rate is 15.4% of periods, compared with 2.7% for the best deployable operational-loss strategy. Ensemble methods offer low-disruption alternatives, while dynamic programming and smoothing serve different governance and execution constraints.

The practical implication is methodological: strategy selection should be evaluated at the forecast-to-plan layer, not only the forecasting layer. Forecast accuracy remains important but should be treated as a diagnostic metric, separate from operational planning utility. The results suggest that some strategy families can improve planning feasibility with limited accuracy loss, while other cases require explicit tradeoffs between stability, inventory exposure, and responsiveness.

Keywords

Demand planning, Forecast-to-plan conversion, Operational feasibility, Planning stability, Execution risk.

1 Introduction

1.1 The Planning-Execution Gap

Demand planning is an operational decision problem, not only a forecasting problem: forecast-driven decisions influence inventory targets, replenishment orders, supplier coordination, warehouse labor, transportation commitments, and planning-system governance. Recent advances in AI-enabled forecasting and time-series foundation models have made more accurate predictions available (Walter et al. 2025; Culot et al. 2024; Petropoulos et al. 2022; Makridakis et al. 2022), but prediction accuracy alone does not guarantee implementation success, because operations execute planning signals, not forecast error metrics. A planner cannot execute WAPE or RMSE; the organization executes an inventory target, a staffing plan, or a replenishment quantity. When forecast-driven plans change faster than supplier lead times, labor cycles, or system change windows can absorb, the forecast and execution layers become misaligned, creating a planning-execution gap in which the operation cannot execute the plan despite its accuracy.

The gap manifests in four operational symptoms:

1. **Execution violations**: Plans change faster than downstream systems can absorb (e.g., warehouse rescheduling, supplier coordination, system updates)
2. **Governance burden**: Each model change requires validation, coordination, and monitoring cycles, delaying adoption
3. **Supplier disruption**: Frequent plan changes force suppliers to expedite orders or renegotiate, increasing costs
4. **Operational gridlock**: Operations teams spend effort coordinating plan changes rather than executing the plan

These symptoms are invisible in accuracy metrics but visible in execution logs, planner workload, and supplier relationships. The operational reality is that forecast accuracy and plan executability are distinct objectives.

1.2 From Forecasting Optimization to Feasibility-Constrained Selection

Forecasting research traditionally optimizes for accuracy, including lower WAPE, lower MAE, and better leaderboard scores. This is valuable, but a more accurate forecast that creates execution gridlock is operationally worse than a slightly less accurate forecast that executes smoothly.

The solution is to shift the evaluation target from *forecast generation* to *forecast-to-plan selection*: ask which planning strategy minimizes operational cost under execution constraints, not only which model minimizes WAPE. This paper operationalizes that shift as a model-agnostic forecast-to-plan feasibility layer: it converts each candidate forecast into an executable planning signal, scores its operational consequence on four metrics (inventory exposure, plan stability, execution violations, governance burden), ranks strategies by feasibility-adjusted utility rather than accuracy alone, and tests ranking stability across execution-risk scenarios and planning contexts.

1.3 Empirical Evidence and Scope

The empirical design uses one execution-risk anchor and three demand-planning datasets. **DataCo** anchors execution-risk scenarios from observed late-delivery heterogeneity; it is not treated as a demand-planning or inventory-control dataset. **Favorita** (main proof of concept) is daily retail demand from 54 stores and 33 product families (1,782 planning units) over 4+ years, testing whether the accuracy-feasibility tradeoff appears in richly-contexted, fine-grained planning. **M5** (hierarchy and intermittency robustness) is daily retail demand from 3,049 items across 10 stores with item-store, department-store, and category-store aggregation, testing whether strategy preferences change with granularity and demand sparsity. **Walmart** is secondary evidence only: a weekly store-department panel with markdown, holiday, and macro signals, used to check whether the same decision layer remains informative under a different cadence and business context. Across 123 paper-facing scenario and robustness evaluation groups, 91.1% show that the WAPE-best strategy differs from the operational-loss-optimal strategy, indicating a structural result rather than a data artifact.

1.4 Objectives

This paper pursues five objectives, each also a research contribution:

- Define a forecast-to-plan feasibility layer separating diagnostic accuracy from operational utility.
- Quantify the planning-execution gap using execution violation rate and normalized planning loss.
- Compare six strategy families under the same feasibility evaluation across data-informed execution-risk scenarios.
- Validate, across Favorita, M5, and secondary Walmart evidence, that strategy rankings shift with scenario, hierarchy, and context.
- Translate the findings into a decision-layer framework for feasibility-aware strategy selection.

The remaining sections follow the same arc as the four symptoms in Section 1.1: Section 3 defines the decision layer, Section 5 reports how often and why accuracy-first and operational-loss-best selection diverge, Section 5.5 organizes the evidence into strategy-selection guidance, and Section 5.6 checks whether the findings remain visible under tighter execution risk, coarser and finer planning grains, and a second dataset with a different cadence and business context.

2 Literature Review

2.1 AI-Enabled Demand Planning and Forecasting Progress

Recent supply-chain research shows rapid progress in AI-enabled demand planning. Walter et al. (2025) identify AI tools, supply-chain applications, and digital transformation as major knowledge clusters; Culot et al. (2024) survey empirical AI adoption and emphasize connecting AI advancement to operational outcomes; Vlachos and Reddy (2025) identify decision-support, execution optimization, and control monitoring as high-value areas. Forecasting research has also broadened beyond single-method accuracy comparisons toward richer practice-oriented evaluation (Petropoulos et al. 2022). Retail benchmarks show the same trend: the M5 competition demonstrates forecasting at scale across hierarchies and intermittency (Makridakis et al. 2022), while tree-based methods remain central in many high-performing retail forecasting pipelines (Januschowski et al. 2022). Recent time-series foundation models further suggest that forecasting pipelines are moving toward broader zero-shot and cross-domain capability (Das et al. 2024; Ansari et al. 2024). These advances make downstream evaluation more urgent: a more accurate forecast is valuable only if operations can execute the resulting plan.

2.2 Forecast-to-Decision and Predict-Then-Optimize Research

Forecast-to-decision research directly addresses why prediction accuracy and downstream decision quality can diverge. Elmachtoub and Grigas (2022) show predict-then-optimize workflows improve by training toward decision loss rather than prediction loss alone; Mandi et al. (2024) survey decision-focused learning, integrating machine learning and constrained optimization to improve decision quality under uncertainty; and Jia et al. (2025) propose scenario predict-then-optimize for online inventory routing, showing routing and replenishment decisions need more than point-forecast accuracy. This paper shares that motivation but a different goal: rather than new forecasting or end-to-end learning algorithms, it proposes an interpretable evaluation layer that lets operations teams compare existing candidate forecasts by operational consequence, not accuracy metrics alone.

2.3 Operations Management and Planning-System Architecture

Advanced planning and ERP systems architecturally separate forecasting, planning, and execution layers (Stadtler et al. 2015), and model deployment requires validation, approval, and integration cycles that take weeks or months. When forecasting models change faster than implementation can absorb, operations teams face a governance problem, not just a technical one. Inventory theory provides the operational foundation: Silver et al. (1998) and Axsater (2015) frame inventory planning as a cost-and-service-level problem rather than a pure prediction problem, Hyndman and Koehler (2006) show forecast-error measures have known limitations, and forecast combination can reduce model-specific swings (Bates and Granger 1969). Forrester (1961) shows that industrial systems amplify instability when material and information flows respond with delays, the bullwhip effect, and the same logic applies here: coordination mechanisms (supplier partnerships, EDI, collaborative planning) reduce the planning-execution gap but do not eliminate it, because governance processes are slow to change compared to forecasting innovation. Rolling-horizon decision theory adds the final piece: Sethi and Sorger (1991) show a locally attractive action can be undesirable once future transitions are considered, which motivates this paper's greedy-to-DP comparison as a measure of planning myopia.

2.4 Research Gap and This Paper's Position

Existing research improves forecasts, combines forecasts, or optimizes downstream decisions in isolation. Fewer studies provide a practical, interpretable decision layer that compares forecasting-driven strategies by their operational consequences, including inventory exposure, plan stability, execution violations, and governance burden, directly at the forecast-to-plan boundary where execution actually happens. This paper addresses that gap; the goal is not to promise a universal deployment rule, but to position forecast accuracy as one signal among several operational objectives, evaluated where planning decisions are executed rather than where forecasts are generated.

3 Methods

3.1 The Planning-Execution Gap: Problem and Design Rationale

Forecasting teams optimize forecast accuracy (WAPE); operations teams execute plans. These objectives often diverge when an improved forecast requires abrupt plan changes that exceed operational capacity, including supplier lead times, warehouse labor constraints, and system change cycles. The paper's solution is to convert all candidate forecasts into a common planning-signal form, apply the same operational metrics to every candidate, and rank strategies by those metrics rather than by WAPE. This ensures fair comparison across strategy families (accuracy-first, ensemble, smoothing, greedy, DP, budgeted-DP) and reveals when accuracy-first selection diverges from operationally-optimal selection.

Figure 1 shows this as a decision-layer framework. Unlike a standard forecasting pipeline (generate forecast $\rightarrow$ report WAPE $\rightarrow$ choose), the layer adds an operational evaluation step: convert forecast to plan, measure feasibility, and compare candidate strategies under the same objective. This figure is *how* the paper measures feasibility; Figure 3 in

Section 5.5 is the complementary *strategy-selection* guide built on those measurements.

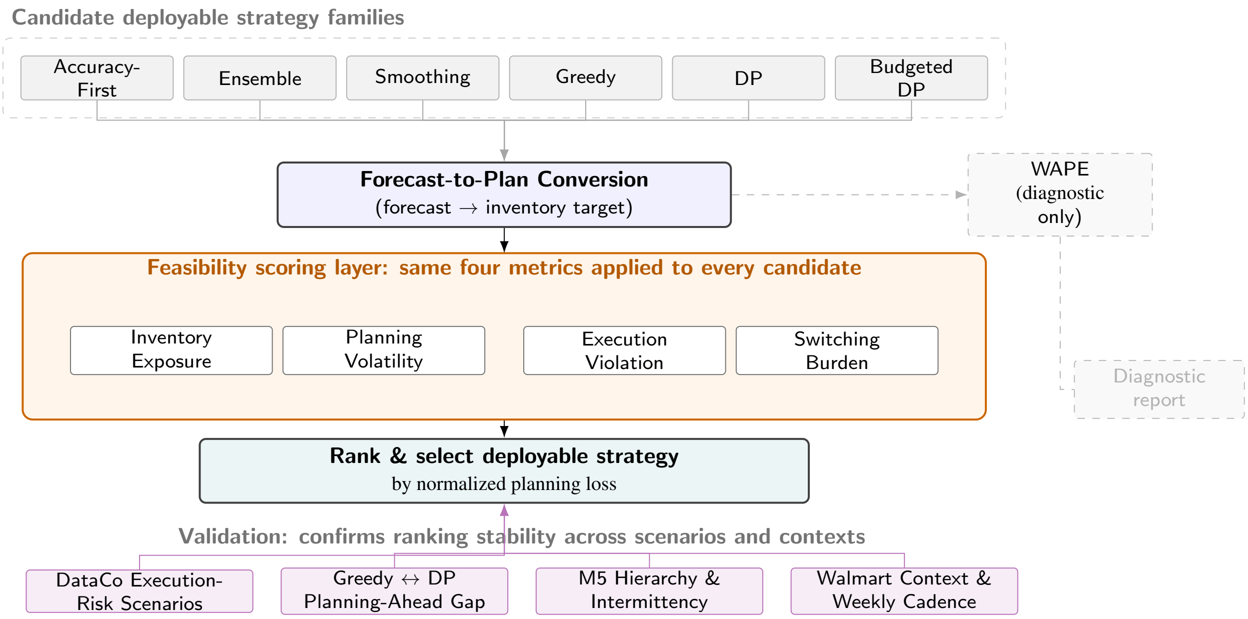

Figure 1. Feasibility-constrained decision layer for forecast-driven demand planning.

3.2 Forecast-to-Plan Conversion: Why and How

The first step converts forecasts into operational plans. Let i index a planning unit, t a planning period, and m a candidate model, so the candidate forecast is $\hat{y}^m_{i,t}$; the conversion is

$$p^m_{i,t} = \hat{y}^m_{i,t} + SS_i, \tag{1}$$

where SS_i is safety stock from demand volatility and service-level targets, and $p^m_{i,t}$ is the resulting inventory target, which is the operational plan.

Different forecasts create different plans: Forecast A might suggest holding 100 units and Forecast B 150, and even if both have similar WAPE, Forecast B requires 50 additional units of inventory and a larger supply-chain adjustment. Inventory targets are the right common form because they are the primary planning signal in demand-driven supply chains, cascading to procurement orders, warehouse allocations, staffing, and transportation. Measuring changes in the target captures operational workload across all of these downstream functions at once.

3.3 Information Access and Leakage Prevention

The evaluation uses three information layers. Calibration uses validation-period realized demand to estimate expected operational costs. Test-period selection does not observe $y_{i,t}$; deployable selectors use candidate forecasts, prior executed plans, execution-capacity scenarios, validation-derived expected costs, switching penalties, and policy rules. Realized test demand is used only after the horizon ends for ex-post scoring. This separation prevents look-ahead bias and distinguishes deployable selectors from the non-deployable oracle references in Section 3.7.

3.4 Feasibility Scoring: Four Operational Metrics

Once all candidates are converted to planning signals, we measure the operational consequence of each. In calibration, these metrics estimate validation-derived expected cost; in ex-post evaluation, they measure realized test-period performance. We use four metrics:

1. Inventory Exposure (Equation 2)

$$IC^m_{i,t} = h[p^m_{i,t} - y_{i,t}]^+ + b[y_{i,t} - p^m_{i,t}]^+ \tag{2}$$

Supply chains balance holding cost h and shortage cost b, revealing whether accuracy improvements hide inventory cost increases. For deployable strategies, this inventory exposure is estimated from validation-period realized demand and used to construct validation-derived expected costs. Test-period realized demand is withheld from selection and used only for ex-post evaluation after the test horizon is complete.

2. Planning Volatility (Equation 3)

$$PV_{i,t}^{m} = |p_{i,t}^{m} - p_{i,t-1}| \tag{3}$$

Operations cannot replan infinitely often: suppliers, warehouses, and systems need stable commitments, so sharp plan changes become coordination bottlenecks. We measure plan movement to quantify this burden.

3. Execution Violation (Equation 4)

$$EV_{i,t}^{m} = [|p_{i,t}^{m} - p_{i,t-1}| - C_{i,t}]^{+} \tag{4}$$

Operations have a maximum absorption rate $C_{i,t}$ per period, set by supplier lead times, labor rescheduling capacity, and system update cycles. When a plan change exceeds $C_{i,t}$, the operation cannot execute it; the excess $EV_{i,t}^{m}$ is the execution violation, and this is what "planning-execution gap" means operationally.

4. Model Switching Burden (implicit in loss function): Each model change requires validation, coordination, and monitoring. These governance processes take time and resources regardless of whether the new model is more accurate; we measure the number of model changes to quantify this burden. Aggregating execution violations over time gives the **planning-execution gap rate**, the fraction of periods where the operation cannot execute:

$$PEG(s) = \frac{1}{T}\sum_{t=1}^{T} \mathbb{I}\{EV_t(s) > 0\}. \tag{5}$$

A strategy with PEG rate 30% requires emergency handling, expediting, or replanning in 30% of periods; PEG 5% is operationally far more feasible.

3.4.1 Why Exclude WAPE from the Objective?

We report WAPE separately as a diagnostic metric rather than folding it into the operational objective: it measures precision, not consequence, so a 10% WAPE improvement that doubles execution violations is operationally worse despite looking better on paper, and it can mask trade-offs since a more accurate forecast often requires more frequent model changes that create governance burden invisible to accuracy metrics. Separating WAPE (diagnostic) from operational metrics (primary) lets planners see when an accuracy improvement is operationally worthless or even harmful.

3.5 Normalized Planning Loss: Combining Operational Metrics

We combine the four operational metrics into a single planning-loss objective:

$$\begin{aligned} L(s) = {} & \lambda_{\text{inventory}} \cdot \mathrm{NIC}(s) + \lambda_{\text{volatility}} \cdot \mathrm{NVC}(s) \\ & + \lambda_{\text{execution}} \cdot \mathrm{NEC}(s) + \lambda_{\text{switch}} \cdot \mathrm{NSC}(s), \end{aligned} \tag{6}$$

where NIC, NVC, NEC, NSC are normalized inventory, volatility, execution, and switching components, and λ values are organizational weights.

Normalization: All components are divided by the accuracy-first baseline (Global Best), so a normalized loss of 0.9 means 10% better than accuracy-first and 1.1 means 10% worse, which remains meaningful even when absolute cost values differ across datasets.

Weight selection: Current weights are $\lambda_{\text{inventory}} = 1.0, \lambda_{\text{volatility}} = 0.10, \lambda_{\text{execution}} = 0.10, \lambda_{\text{switch}} = 0.05$, reflecting a supply chain where inventory cost dominates but feasibility and governance still matter; organizations can adjust these by constraint, e.g., raising $\lambda_{\text{execution}}$ under tight execution capacity or λ_{switch} under strict governance.

3.6 From Metrics to a Selection Procedure

Figure 1 assembles the equations above into one procedure. Candidate strategies generate forecasts, convert them into planning signals (Equation 1), and pass through the same feasibility-scoring layer. Validation-derived expected inventory exposure, planning volatility, execution violation, and switching burden are computed under one operational objective (Equations 2 to 5). These scores are aggregated into $L(s)$ (Equation 6), ranking deployable strategies without test-period realized demand. Ex-post validation then checks realized test outcomes, execution-risk scenarios, oracle benchmarks, and cross-dataset robustness.

The main methodological contribution is the shared feasibility-scoring layer and its integration with multiple deployable strategy families. By routing accuracy-first, ensemble, smoothing, greedy, DP, and budgeted-DP policies through the same operational objective, the framework makes strategy families comparable as planning decisions rather than as isolated forecasting methods. The layer is agnostic to strategy family; we evaluate six. **Accuracy-First** chooses the model with lowest validation WAPE (the conventional benchmark). **Ensemble** combines candidate forecasts by simple averaging or operational-loss weighting, dampening model-specific swings while retaining both models' information. **Smoothing** blends the new plan with the previous period's,

$$p_t = \alpha p_t^{\text{candidate}} + (1-\alpha) p_{t-1}, \tag{7}$$

where lower α slows adaptation and reduces execution burden, trading stability for responsiveness. **Greedy** selection minimizes only the current-period cost $c_t(m_{t-1}, m)$ of switching from model m_{t-1} to m,

$$m_t^G = \arg\min_{m \in \mathscr{M}} c_t(m_{t-1}, m), \tag{8}$$

ignoring future consequences. **Dynamic Programming** corrects this by accounting for future transitions,

$$V_t(m_{t-1}) = \min_{m \in \mathscr{M}} \{c_t(m_{t-1}, m) + V_{t+1}(m)\}, \quad V_{T+1}(\cdot) = 0; \tag{9}$$

the greedy-to-DP gap measures the cost of this myopia, small when switching cost is low and material when coordination costs are high. **Budgeted DP** adds a hard governance constraint on total switches,

$$\sum_{t=2}^{T} \mathbb{I}\{m_t \neq m_{t-1}\} \leq K, \tag{10}$$

relevant when an organization imposes a fixed annual limit on model changes.

3.7 Validation Benchmarks: Non-Deployable Oracles

Two oracle benchmarks provide diagnostic references, neither deployable. Only the Realized-Inventory Oracle DP and the perfect-demand diagnostic reference use test-period realized outcomes during selection; both are explicitly non-deployable diagnostic references. The **Realized-Inventory Oracle DP** uses realized test-period inventory outcomes over the same forecast candidates, information no real planner has at decision time. The DP-oracle gap is computed as each strategy's normalized loss minus the Realized-Inventory Oracle DP's normalized loss within the same evaluation group, so it measures selection suboptimality relative to this hindsight benchmark. The **perfect-demand diagnostic reference** uses realized demand itself as the planning signal, but it should not be treated as a guaranteed lower bound: exact demand tracking can require plan changes faster than execution capacity allows, producing high execution violations. This is an *accuracy-induced feasibility failure*: a forecast that tracks demand perfectly can still yield an infeasible plan, and its reference gap is therefore diagnostic only, combining forecast uncertainty and feasibility effects to show that accuracy alone is insufficient.

4 Data Collection and Experimental Design

The empirical design uses one execution-risk anchor and three demand-planning datasets, summarized in Table 1. DataCo is not an inventory-control dataset; it transfers empirical heterogeneity in late-delivery risk into scenario design. Favorita is the main forecast-to-plan proof of concept, M5 tests hierarchy and intermittent-demand robustness, and Walmart provides secondary business-context and weekly-cadence robustness. Table 1 is therefore a validation design rather than a source inventory: each row states the research question, technical test, data source, and thesis role.

Table 1. Empirical validation design for feasibility-constrained model selection.

Evidence block	Research question	Technical test	Data source	Thesis role
Execution-risk anchoring	Is execution risk measurable from real operations data?	Late-delivery percentile scenarios and execution-weight mapping.	DataCo	Calibrates scenario severity; not a demand dataset.
Forecast-to-plan proof of concept	Does WAPE-best differ from feasibility-best?	Forecast-to-plan conversion and normalized planning loss comparison.	Favorita	Main empirical proof of the planning-execution gap.
Strategy-family mechanism	Why do ensemble, smoothing, and DP differ?	Shared operational scoring for deployable strategy families.	Favorita	Explains mechanism-level tradeoffs.
Hierarchy and intermittency robustness	Does planning grain change strategy choice?	Item-store, department-store, category-store, and intermittency checks.	M5	Tests planning-grain robustness.
Business-context robustness	Does weekly cadence and context alter strategy choice?	History-only vs. context-aware forecasts, cadence, and switch-budget tests.	Walmart	Secondary robustness for operating context.

DataCo execution-risk anchor. Supply-chain fulfillment records with late-delivery outcomes. Late-delivery anchors from context-level percentiles range from 0.535 (low-risk) to 0.901 (severe-risk); under the additive clipped scenario mapping, these produce execution weights from 0.318 to 0.500. These are sensitivity weights, not exact dollar costs.

Favorita proof of concept. A daily retail demand-planning panel from Corporacion Favorita treating each store-family pair as a planning unit, with promotion, holiday, oil-price, transaction, and store-context information. Candidate forecasts include transparent baselines and, when available, LightGBM and XGBoost (Ke et al. 2017; Chen and Guestrin 2016).

M5 robustness design. M5 contains 3,049 items across 10 stores (Makridakis et al. 2022); the analysis samples item-store series and evaluates item-store, department-store, and category-store planning grains and intermittent-demand buckets, testing whether the planning-execution gap persists when demand is sparse or aggregated.

Walmart business-context robustness design. Weekly store-department sales with holiday, markdown, store, and economic context fields, used as a secondary robustness check. It compares history-only against history-plus-context forecasts under holiday/markdown stress windows, weekly cadence constraints, and switch-budget sensitivity, testing whether the feasibility layer stays informative when planning is weekly and context-driven rather than daily. Additional local-store retail datasets are left for future robustness work.

5 Results and Discussion

5.1 Finding 1: The Planning-Execution Gap is Measurable and Structural

The analysis begins with a fundamental question: when accurate forecasts and operational feasibility diverge, how often does this matter? In 91.1% of 123 paper-facing scenario and robustness evaluation groups, the strategy that minimizes WAPE is not the strategy that minimizes operational planning loss. Accuracy-first strategies have a PEG rate of 15.4% of periods, compared with 2.7% for the best deployable operational-loss strategy, a 12.7-percentage-point gap. The planning-execution gap is therefore measurable and operationally consequential.

When accuracy-best and operational-loss-best strategies diverge, no single method is universally optimal; different strategy families serve different operational constraints, starting with ensemble methods.

5.2 Finding 2: Ensemble Methods Provide Low-Disruption Alternatives

Ensemble averaging dampens abrupt plan movement while retaining information from multiple models. In the Favorita baseline evaluation, Global Best is the accuracy-first reference with WAPE 10.9%, violation rate 28.4%, and normalized loss 1.550. Operational Ensemble has WAPE 11.9%, violation rate 13.3%, and loss 1.422; Simple Ensemble has WAPE 12.9%, violation rate 6.6%, and loss 1.434. Ensemble is therefore practical when a one- to two-percentage-point WAPE loss is acceptable in exchange for lower execution burden.

Figure 2 places all nine baseline-scenario strategies on the same accuracy-execution plane. WAPE and execution penalty do not move together: Global Best and Family Best sit near the strongest WAPE region, while ensemble policies accept higher WAPE for much lower execution penalty. The perfect-demand diagnostic reference has zero forecast error but the highest execution penalty, illustrating the accuracy-induced feasibility failure introduced in Section 3.

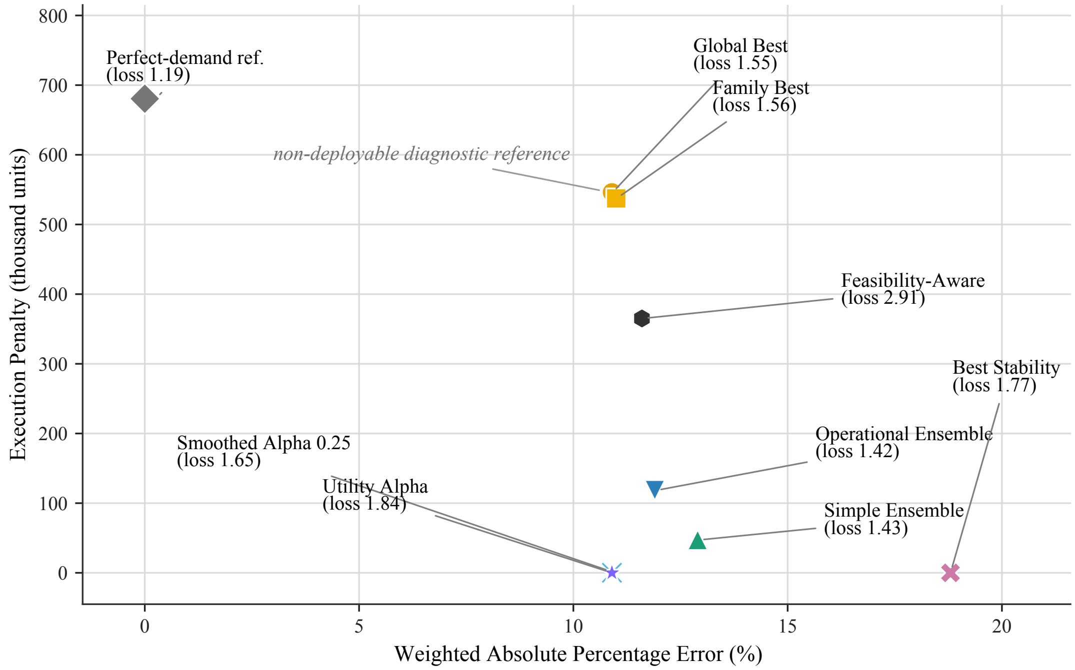

Figure 2. Favorita baseline accuracy-execution frontier; the zero-WAPE marker is a non-deployable perfect-demand diagnostic reference.

5.3 Finding 3: Smoothing and DP Serve Different Governance Regimes

When ensemble's improvement is insufficient, smoothing and dynamic programming cover the remaining operational constraints. **Smoothing** slows adaptation by blending the new plan with the previous period's, trading response speed for stability. In the Favorita baseline, Smoothed Alpha 0.25 retains WAPE 10.9% and reduces the execution violation rate to 0.1%, but its normalized loss rises to 1.646 because inventory exposure increases. The Best Stability reference reaches zero violations with WAPE 18.8% and normalized loss 1.766. Smoothing is therefore valuable where execution capacity is tightly constrained, but it is not a free improvement; the tradeoff with inventory exposure and responsiveness must be explicit.

Dynamic programming accounts for future switching costs that one-period greedy selection ignores: staying with an incumbent model can beat switching to a marginally better one once future coordination cost is counted. The greedy-to-DP gap (0.044, small when switching cost is low) and DP-oracle gap (0.198) quantify this myopia and the feasibility remaining after DP selection; DP is worth its complexity when changes require multi-step coordination or system synchronization, and budgeted DP adds a hard annual switching limit under strict change-control governance.

5.4 Finding 4: Planning Hierarchy Changes Optimal Strategy

The most striking finding is that different aggregation levels favor fundamentally different strategies, challenging the "one global forecasting model" paradigm. In the M5 hierarchy evaluation, the finest grain (item-store, sparse and volatile) favors Smoothed Alpha 0.25 (WAPE 36.8%, violation 1.0%, loss 1.223) because smoothing dampens noise without model changes; aggregating to department-store demand shifts the advantage to Operational Ensemble (WAPE 16.2%, violation 10.0%, loss 1.177), and category-store aggregation keeps it (WAPE 15.9%, violation 6.3%, loss 1.261), since ensembles can capture different models' information once demand is more stable. The evidence argues against treating one global model as uniformly preferred across all planning grains: item-level favors smoothing, department- and category-level favor ensemble, with DP added where governance matters. Planning-system architecture should be hierarchy-aware for exactly this reason.

5.5 Strategy Selection Under Operational Constraints

Findings 1 to 4 above are diagnostic: they explain *why* strategies diverge. This section turns that diagnosis into strategy-selection guidance for comparing candidate forecasts as operational planning decisions.

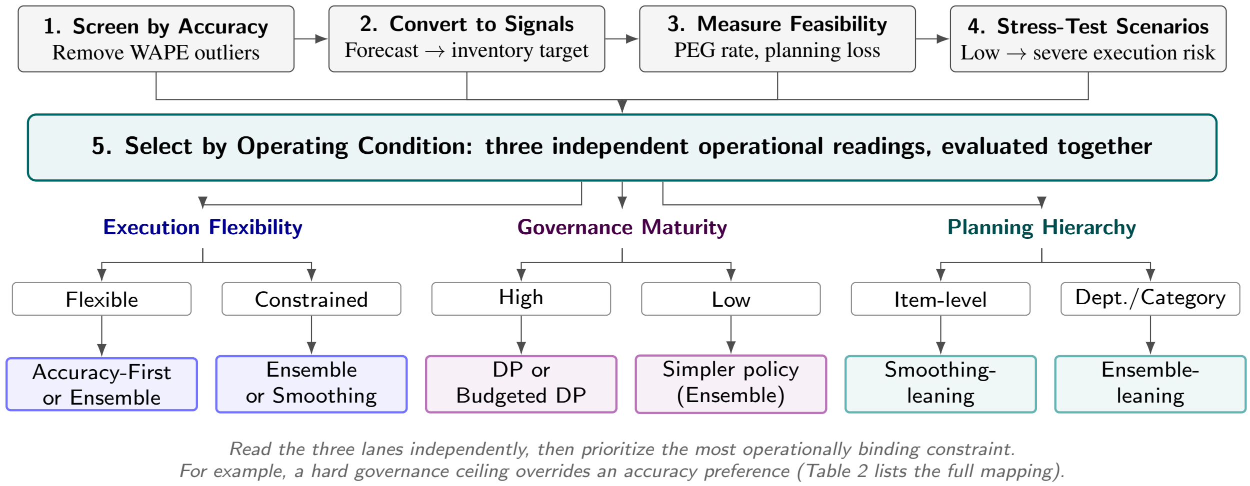

Figure 3. Decision-layer guidance for strategy selection under operational constraints.

Steps 1 to 4, shown across the top of Figure 3, screen out weak forecasts, convert survivors into inventory targets, measure PEG rate and normalized planning loss, and stress-test candidates across execution-risk scenarios. Step 5 combines the three operational readings established by the findings: execution flexibility (Findings 1 and 2), governance maturity (Finding 3), and planning hierarchy (Finding 4). Table 2 gives the condition-to-policy mapping behind each leaf.

Table 2. Strategy families and operating conditions.

Operating condition	Policy	Reason
Flexible execution capacity	Accuracy-first or ensemble	WAPE gains are useful when plan movement is absorbable, but feasibility screening is still required.
Moderate execution constraint	Ensemble	Averaging dampens model-specific swings while retaining adaptation.
Severe execution constraint	Smoothing	Gradual adaptation reduces execution burden when abrupt movement is costly.
High switching governance	DP or budgeted DP	Horizon-aware selection accounts for future switching and coordination burden.
Multi-level hierarchy	Choose by planning grain	Item, department, and category planning favor different mechanisms.

The framework is structured evaluation, not single-method prescription; the final decision still depends on organizational constraints and risk tolerance. Section 5.6 checks whether the evidence remains consistent outside the main Favorita proof of concept.

Using this decision layer requires the three organizational capabilities in Table 3. Outcomes should be tracked across forecasting, planning, and operational metrics so feasibility tradeoffs are visible to the teams who select and execute planning changes, not only to the team that built the forecast.

Table 3. Implementation agenda for a feasibility-aware planning system.

Capability	Key actions
Data & measurement	Estimate execution capacity from lead times and change cycles; log exception periods; calibrate violation cost.
Governance design	Define selection thresholds for WAPE, violation rate, and planning loss; stress-test before selection; align forecasting and operations on shared criteria.
System integration	Convert forecast-to-plan at the system boundary; embed capacity constraints in optimization; monitor violations and loss alongside WAPE.

5.6 Robustness and Validation

Four complementary checks examine whether the findings are structural rather than data artifacts and whether the guidance in Section 5.5 is consistent outside the conditions it was built on.

5.6.1 Baseline and External Validation: Is This a Real Problem?

The 91.1% mismatch rate and 12.7pp PEG-rate gap demonstrate that WAPE-best and operationally best strategies answer different questions. This divergence is consistent with predict-then-optimize research showing that prediction-loss-minimizing models need not minimize downstream decision loss (Elmachtoub and Grigas 2022; Jia et al. 2025). This paper contributes a feasibility-specific metric, PEG rate, and an evaluation layer for demand planning.

5.6.2 Scenario Validation: Does Execution Pressure Change Rankings?

DataCo-informed execution-risk scenarios validate whether tighter execution constraints change strategy rankings. As execution-risk weights increase from low to severe, ensemble and smoothing methods become relatively more attractive than accuracy-first selection, supporting scenario testing before strategy selection.

Figure 4 orders the five scenarios by increasing severity and tracks all nine strategies through each; the y-axis is broken because Feasibility-Aware, an early naive selector, sits nearly twice as high as everyone else and would otherwise compress the eight competitive strategies into an unreadable band. Among those eight, Global Best and Family Best rise fastest as execution risk tightens (1.55 at baseline to 2.00 at severe), while Simple Ensemble, Operational Ensemble, and the smoothing-based strategies stay comparatively flat. This visual pattern supports the interpretation that accuracy-first selection is especially exposed to execution-risk escalation.

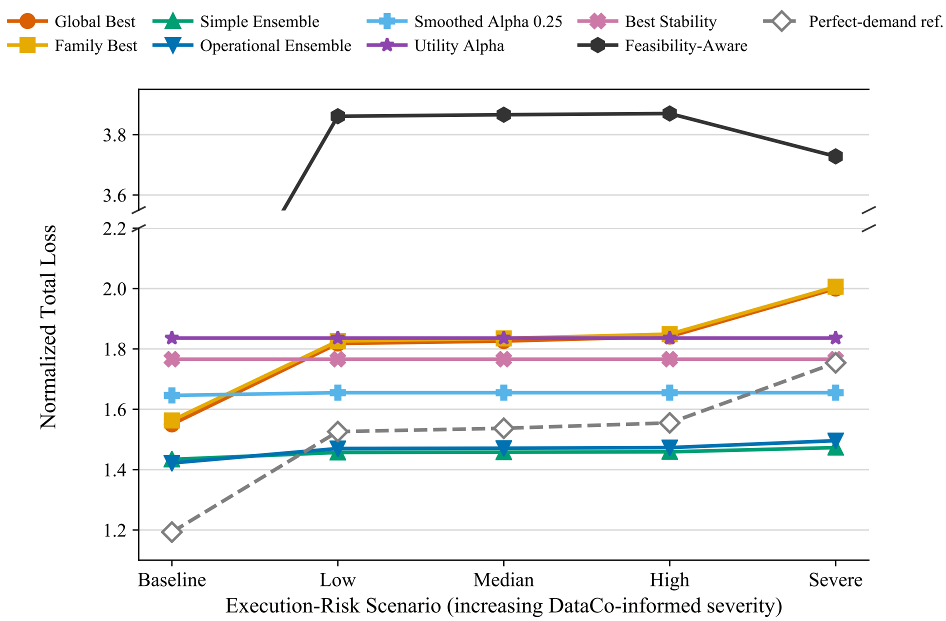

Figure 4. Favorita normalized planning loss across execution-risk scenarios, ordered by increasing severity. The broken y-axis separates Feasibility-Aware (top panel) from the eight competitive strategies (bottom panel).

Algorithmic validation asks whether planning ahead has value: the greedy-to-DP gap (0.044) measures the cost of myopic one-period selection. DP is therefore a deliberate choice for high-governance environments, not a universal improvement.

5.6.3 Cross-Dataset Robustness Validation

Favorita is the main proof of concept; M5 shows that hierarchy and sparsity change the optimal strategy; Walmart provides secondary evidence in a weekly-cadence setting, with an 80.0% accuracy-first mismatch rate, suggesting that the gap is not limited to daily retail planning. Together, the checks support the thesis that feasibility-constrained planning is a multi-objective tradeoff surface, not a single-method leaderboard. Baseline validation establishes the gap, scenario and algorithmic validation support the execution-flexibility and governance lanes, and cross-dataset validation supports the planning-hierarchy lane.

5.7 Limitations and Future Work

Public datasets do not observe the full internal cost of execution changes, planner overrides, or system reconfiguration, so execution capacity is proxied rather than directly measured; internal operational data could improve this calibration. The study also focuses on model-selection strategy rather than real-time plan adjustment or human planner intervention. Extending the framework to planner workflows and collaborative replanning is a natural next step.

6 Conclusion

This paper reframes operational demand planning as feasibility-constrained model selection through a forecast-to-plan evaluation layer. The central finding is that in 91.1% of 123 paper-facing scenario and robustness evaluation groups, the accuracy-best deployable strategy diverges from the operational-loss-optimal deployable strategy. The accuracy-first execution violation rate (15.4% of periods) is about 5.7 times higher than the best deployable operational-loss strategy (2.7% of periods).

Section 1.1 opened with four operational symptoms. *Execution violations* are captured by PEG rate. *Governance burden* is addressed by DP or budgeted DP where switching requires coordination. *Supplier disruption* appears through volatility and execution violations, where ensembles and smoothing can reduce burden but may require explicit trade-offs. *Operational gridlock* is addressed by replacing ad hoc negotiation with a repeatable, auditable strategy-selection procedure. Together, the metrics and strategy-family results connect operational symptoms to measurable decision criteria and deployable strategy choices.

This work bridges forecasting advancement and operational execution: forecast accuracy is necessary but not sufficient for operational use. Making execution feasibility measurable and comparable across strategies lets planners evaluate accuracy, stability, and operational realism together. The goal is not to replace forecasting metrics but to evaluate them where planning decisions are actually executed.

Biography

Yihua Xu is an independent researcher and AI and Data professional based in San Jose, California, USA. She currently works at Accenture in the AI and Data practice, with professional interests spanning machine learning, operations research, supply chain analytics, and decision intelligence. She received her Bachelor of Science degree in Industrial Engineering from the H. Milton Stewart School of Industrial and Systems Engineering at the Georgia Institute of Technology and her Master of Business Analytics degree from the MIT Sloan School of Management at the Massachusetts Institute of Technology. Her research focuses on feasibility-aware decision frameworks for operational planning systems, including demand planning, forecast-to-plan conversion, planning stability, execution-risk evaluation, and reverse logistics. Her work aims to connect predictive models with executable operational decisions under inventory, capacity, and planning-governance constraints.

References

Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S. S., Pineda Arango, S., Kapoor, S., Zschiegner, J., Maddix, D. C., Wang, H., Mahoney, M. W., Torkkola, K., Wilson, A. G., Bohlke-Schneider, M. and Wang, Y., Chronos: Learning the language of time series, *Transactions on Machine Learning Research*, October 2024.

Axsater, S., *Inventory Control*, 3rd Edition, Springer, 2015.

Bates, J. M. and Granger, C. W. J., The combination of forecasts, *Operational Research Quarterly*, vol. 20, no. 4, pp. 451-468, 1969.

Chen, T. and Guestrin, C., XGBoost: A scalable tree boosting system, *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785-794, 2016.

Culot, G., Podrecca, M. and Nassimbeni, G., Artificial intelligence in supply chain management: A systematic literature review of empirical studies and research directions, *Computers in Industry*, vol. 162, article 104132, 2024.

Das, A., Kong, W., Sen, R. and Zhou, Y., A decoder-only foundation model for time-series forecasting, *Proceedings of the 41st International Conference on Machine Learning*, PMLR, vol. 235, pp. 10148-10167, 2024.

Elmachtoub, A. N. and Grigas, P., Smart predict, then optimize, *Management Science*, vol. 68, no. 1, pp. 9-26, 2022.

Forrester, J. W., *Industrial Dynamics*, MIT Press, Cambridge, MA, 1961.

Hyndman, R. J. and Koehler, A. B., Another look at measures of forecast accuracy, *International Journal of Forecasting*, vol. 22, no. 4, pp. 679-688, 2006.

Januschowski, T., Wang, Y., Torkkola, K., Erkkila, T., Hasson, H. and Gasthaus, J., Forecasting with trees, *International Journal of Forecasting*, vol. 38, no. 4, pp. 1473-1481, 2022.

Jia, M., Schrotenboer, A. H. and Chen, F., Scenario predict-then-optimize for data-driven online inventory routing, *Transportation Science*, vol. 59, no. 5, pp. 1032-1056, 2025.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T., LightGBM: A highly efficient gradient boosting decision tree, *Advances in Neural Information Processing Systems*, vol. 30, pp. 3146-3154, 2017.

Makridakis, S., Spiliotis, E. and Assimakopoulos, V., M5 accuracy competition: Results, findings, and conclusions, *International Journal of Forecasting*, vol. 38, no. 4, pp. 1346-1364, 2022.

Mandi, J., Kotary, J., Berden, S., Mulamba, M., Bucarey, V., Guns, T. and Fioretto, F., Decision-focused learning: Foundations, state of the art, benchmark and future opportunities, *Journal of Artificial Intelligence Research*, vol. 80, pp. 1623-1701, 2024.

Petropoulos, F. et al., Forecasting: Theory and practice, *International Journal of Forecasting*, vol. 38, no. 3, pp. 705-871, 2022.

Sethi, S. P. and Sorger, G., A theory of rolling horizon decision making, *Annals of Operations Research*, vol. 29, no. 1-4, pp. 387-416, 1991.

Silver, E. A., Pyke, D. F. and Peterson, R., *Inventory Management and Production Planning and Scheduling*, 3rd Edition, John Wiley and Sons, New York, 1998.

Stadtler, H., Kilger, C. and Meyr, H., *Supply Chain Management and Advanced Planning: Concepts, Models, Software, and Case Studies*, 5th Edition, Springer, 2015.

Vlachos, I. and Reddy, P. G., Machine learning in supply chain management: Systematic literature review and future research agenda, *International Journal of Production Research*, vol. 63, no. 16, pp. 5987-6016, 2025.

Walter, A., Ahsan, K. and Rahman, S., Application of artificial intelligence in demand planning for supply chains: A systematic literature review, *The International Journal of Logistics Management*, vol. 36, no. 3, pp. 672-719, 2025.